
WorldDeciding

Final Defense Presentation

Thesis summary, architecture, implementation, evaluation, limitations, and future work.

Design and Development of a Web Platform for Cross-Cultural Opinion Analysis

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Agenda

- 1 Problem and objective
- 2 System design and implementation
- 3 Security, privacy, and evaluation
- 4 Limitations and future work
- 5 Final contribution

Thesis Problem

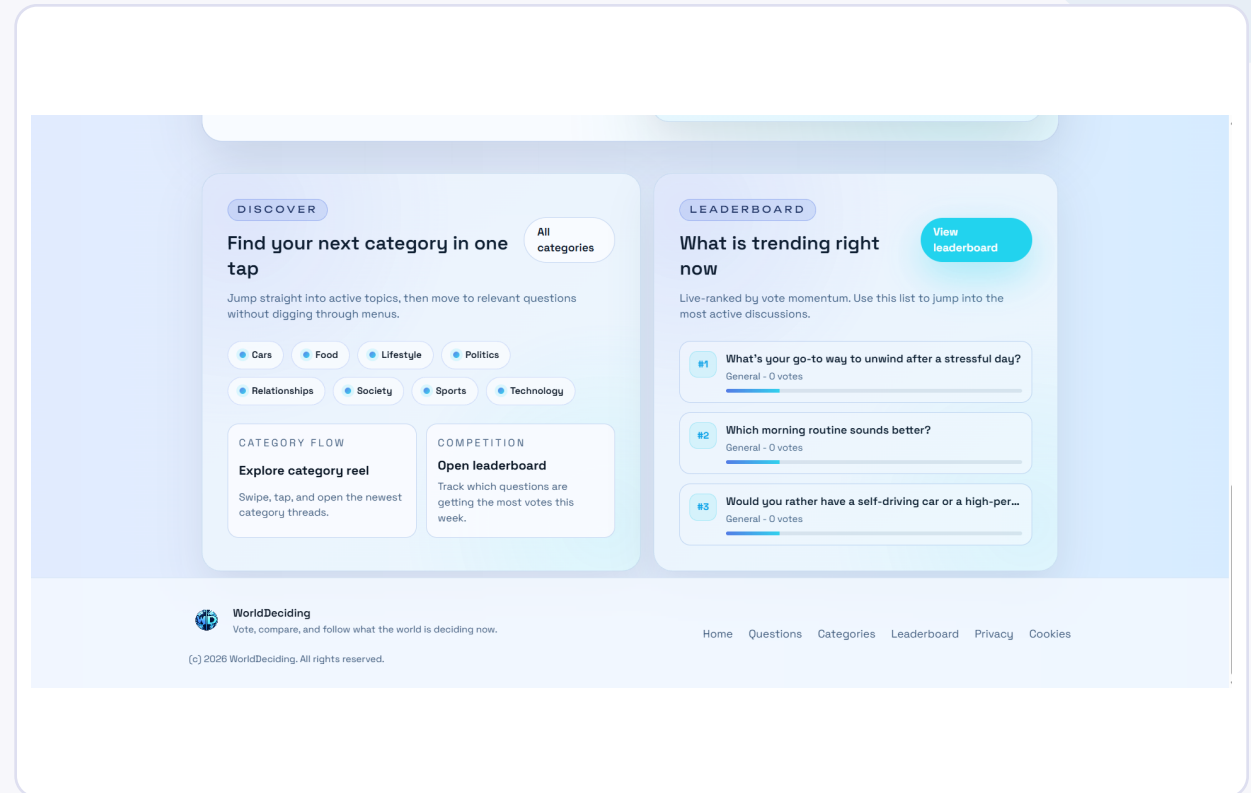
- People express opinions online, but simple totals hide cultural variation.
- Public voting systems need authentication, spam protection, privacy, and clear statistics.
- The goal is to design and develop a practical platform for cross-cultural opinion analysis.

Research and Engineering Objective

- Collect structured votes and qualitative comments.
- Compare opinions by country, age band, and gender.
- Use AI only as an explanatory assistant, not as a decision maker.
- Implement the system using modern full-stack technologies.

System Overview

- React/Vite SPA communicates with ASP.NET Core Web API.
- MediatR handlers implement application use cases.
- PostgreSQL stores durable relational data.
- Redis supports cache and rate limiting.
- Docker Compose runs the required services.



Contribution Map

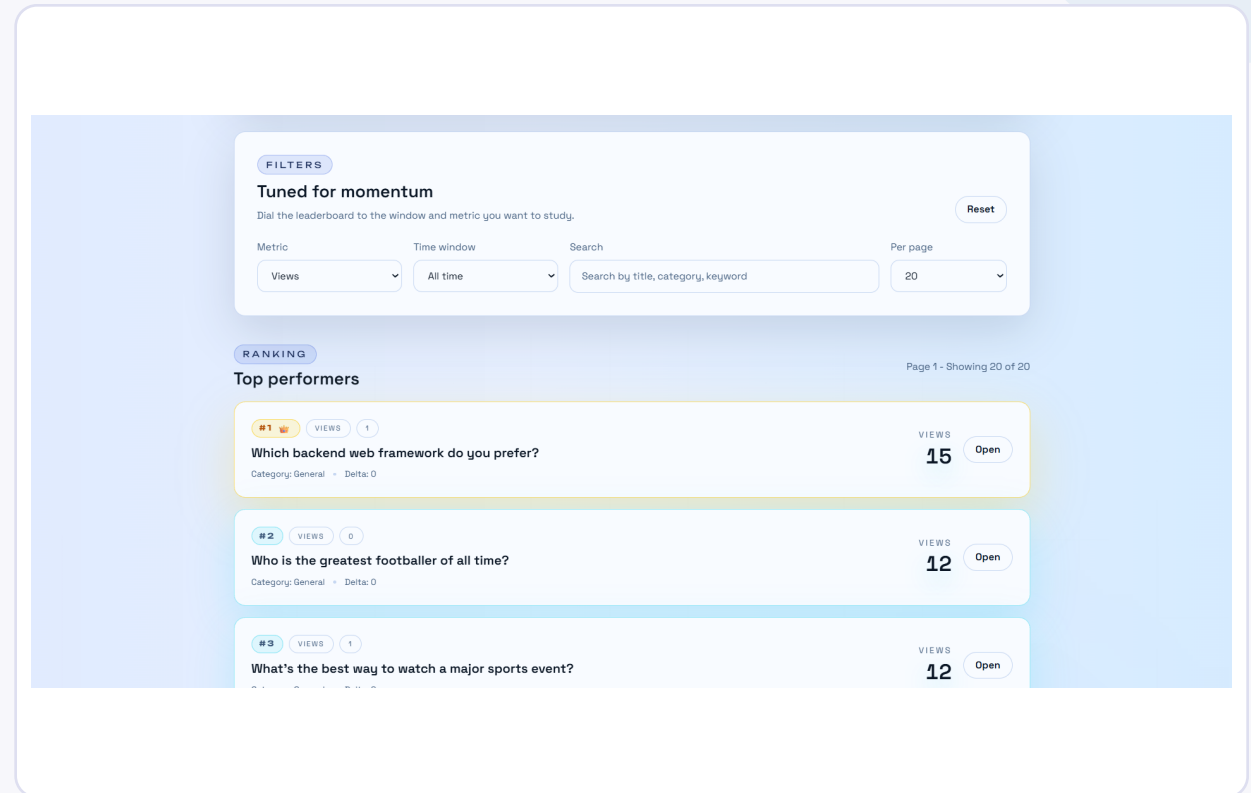
Contribution	Implementation Evidence	Academic Value
Voting platform	Question, option, vote, user, and cooldown logic.	Structured opinion collection.
Comparative statistics	Country, option, age, gender, and global baseline views.	Cross-cultural interpretation support.
Discussion analysis	Comments, likes, replies, and AI summaries.	Combines quantitative and qualitative signals.
Responsible design	Rate limits, IP hashing, consent, and admin roles.	Privacy-aware engineering prototype.

Core Contribution 1: Voting

- Authenticated users can vote on binary or multi-option questions.
- Database uniqueness enforces one active vote per question per user.
- Vote changes are allowed after a 24-hour cooldown.
- Country metadata is stored for comparison, while raw IP is not persisted.

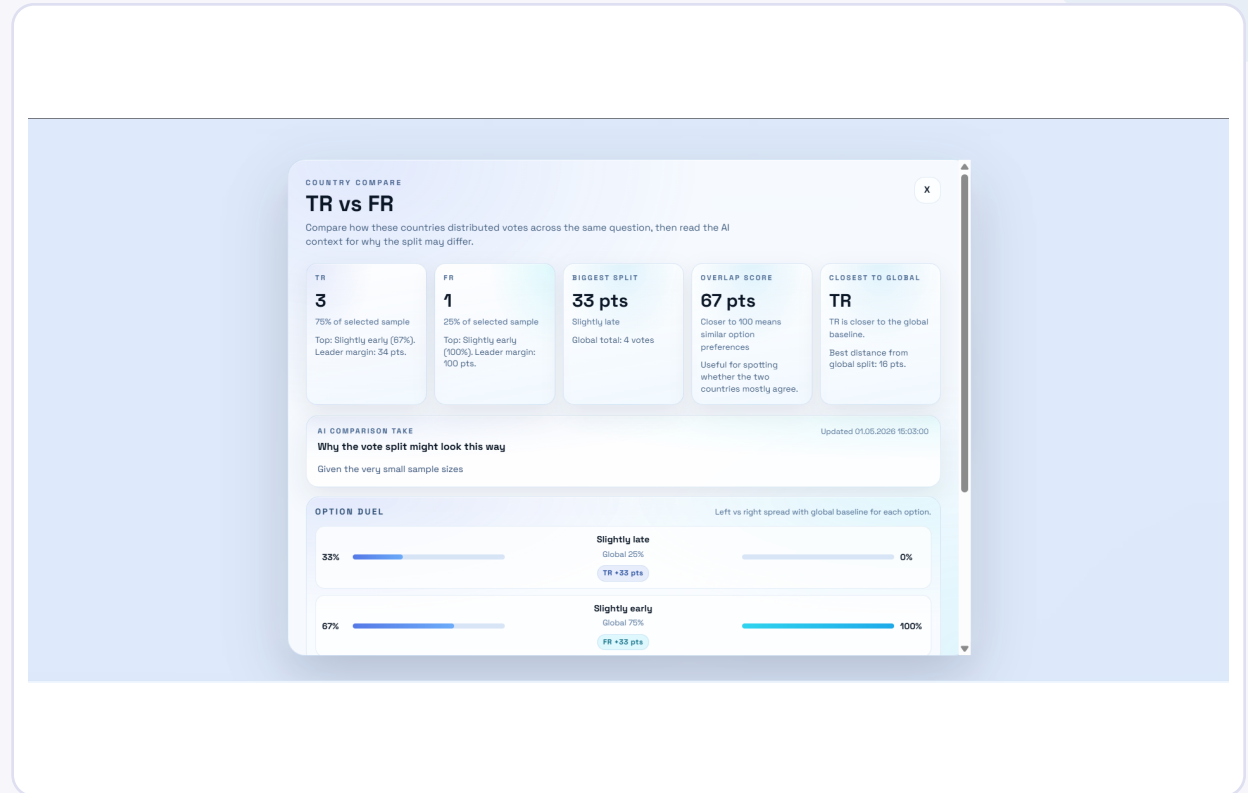
Core Contribution 2: Statistics

- Option distribution shows count and percentage.
- Country distribution enables geographic comparison.
- Demographic breakdowns use user profile fields.
- Leaderboard uses daily aggregated views and votes.



Core Contribution 3: Cross-Country Analysis

- Country comparison shows two national vote patterns and a global baseline.
- The UI highlights biggest split, overlap score, and closest-to-global signal.
- AI text uses tentative language and fallback behavior.
- Small-sample suppression is identified as a future improvement.



Security and Privacy

- Identity password rules, email confirmation, JWT validation, and refresh-token rotation.
- Role-based admin authorization and CORS allowlisting.
- Redis-backed abuse counters for login, forgot password, voting, views, and AI summary.
- Salted IP hashes, analytics consent, and security response headers.

Quality Review

Verified by Documentation

- Screenshots demonstrate completed frontend flows.
- Architecture and data model are described in the thesis.
- Installation PDF lists environment and run steps.
- User guide explains role-specific usage.

Residual Engineering Risk

- Test coverage remains limited.
- Production monitoring is not yet configured.
- Data export and deletion workflows need extension.
- Small sample protections should be enforced in UI and API.

Evaluation

- Functional review confirms implementation of the planned flows.
- Screenshots demonstrate completed user and admin interfaces.
- Build verification should be run before submission.
- Test coverage is the main engineering weakness and should be expanded.

Limitations

- Results are platform participation signals, not representative national statistics.
- GeoIP can be inaccurate under VPNs, proxies, or mobile routing.
- AI summaries can miss nuance and must remain labeled.
- Comment moderation and minimum-threshold privacy suppression need production work.

Future Work Roadmap

1

Data Quality

Weighted sampling, confidence intervals, and sample-size warnings.

2

Safety

Moderation, reporting, audit logging, and admin review queues.

3

Testing

Automated unit, integration, and browser workflow tests.

4

Operations

Secret management, CI/CD, backups, monitoring, export, and deletion.

Final Claim

WorldDeciding is a complete full-stack prototype for exploratory cross-cultural opinion analysis.

- It combines public voting, comparison, comments, AI-assisted summaries, admin tooling, and privacy controls.
- Its results must be interpreted as platform participation signals, not representative national statistics.
- The design is extensible enough for future research and production hardening.

Conclusion

- WorldDeciding implements a complete full-stack prototype for cross-cultural opinion analysis.
- The project combines voting, statistics, discussion, AI assistance, security controls, and admin tooling.
- The thesis contribution is the design and development of a practical, extensible, privacy-aware opinion platform.